CUSTOMER SUPPORT COPILOT

RAG Architecture Comparison and Evaluation Report

Dandes Electronics Knowledge Base | Evaluation date: 31 October 2026
Evaluation set: 20 questions per RAG version (60 answers total)

1. Executive Summary

This project compares V1 Basic RAG, V2 Single-Index RAG, and V2 Multi-Index RAG using the same 20-question evaluation set. Based on the latest recorded evaluation, Multi-Index achieved the highest Recall@5 (94.38%), Coverage (100.00%), and Routing Accuracy (95.00%). It also achieved 100.00% refusal accuracy with zero false refusals. Its Faithfulness (LLM Judge) result is 12.50%, indicating answer grounding remains an important area for improvement.

2. System Versions

V1 Basic RAG: Baseline retrieval and answer generation using the shared knowledge base.

V2 Single-Index RAG: Metadata filtering, hybrid search, reranking, and query transformation.

V2 Multi-Index RAG: Routes questions to specialized knowledge indexes and handles out-of-scope queries through a configured refusal.

3. Retrieval and Refusal Metrics

Metric	V1 Basic	V2 Single	V2 Multi
Precision@5	28.75%	30.00%	28.75%
Recall@5	91.25%	92.50%	94.38%
F1@5	41.84%	43.09%	42.14%
Coverage	93.75%	93.75%	100.00%
Routing Accuracy	N/A	N/A	95.00%
Refusal Accuracy	95.00%	95.00%	100.00%
False Refusals	1	1	0
Questions Evaluated	20	20	20

Interpretation: V2 Multi-Index leads on recall, coverage, routing, and refusal handling. V2 Single-Index has the highest Precision@5 and F1@5. These results show trade-offs rather than a single winner.

4. Generation Evaluation

Generation evaluation currently uses custom LLM-as-Judge faithfulness classification and refusal evaluation. Faithfulness is the percentage of evaluated non-refusal answers classified as *Faithful*; *Partial* and *Unfaithful* are not counted as faithful. Refusal accuracy measures correctly handled questions across the full set.

Metric	V1 Basic	V2 Single	V2 Multi
Faithfulness (LLM Judge)	100.00%	73.33%	12.50%
Refusal Accuracy	95.00%	95.00%	100.00%
False Refusals	1	1	0
Questions Evaluated	20	20	20

Interpretation: V1 has the highest Faithfulness (LLM Judge) percentage. V2 Multi-Index has lower faithfulness despite stronger retrieval and routing results. Review generated answers against retrieved context to identify grounding issues.

5. V2 Multi-Index Target Assessment

Target	Recorded Result	Assessment
Routing Accuracy ≥ 85%	95.00%	Met
Recall@5 ≥ 0.80	94.38%	Met
Faithfulness (LLM Judge) ≥ 85%	12.50%	Not met
Refusal Accuracy = 100%	100.00%	Met
False Refusals = 0	0	Met

6. Limitations and Recommended Improvements

• Improve answer grounding by requiring claims to be supported by retrieved evidence and abstaining when evidence is insufficient.

• Inspect low-faithfulness responses against retrieved passages and expected answers.

• Investigate multi-index routing, ranking, and context assembly for cases with missed or incomplete evidence.

• Confirm the golden evaluation set, reference answers, expected source IDs, and expected indexes match the assignment specification.

• Report custom LLM-as-Judge metrics separately from other evaluation frameworks if those are run in a future evaluation.

7. Reproducibility and Artifacts

The evaluation directory contains the golden test set and evaluation scripts. Per-version results, retrieval metrics, generation metrics, and the consolidated comparison table are stored in evaluation/results/ and evaluation/comparison_table.csv. Demo output is stored in demo_output.txt.

Keep API credentials in the local .env file. Do not include it in a public repository or submission archive. Refer to README.md for execution instructions.

Conclusion

The latest recorded results support V2 Multi-Index RAG for retrieval recall, coverage, routing, and refusal handling. However, its Faithfulness (LLM Judge) result is substantially below the stated target. The system demonstrates promising retrieval and routing performance, while answer grounding requires further investigation and improvement.